
Offline Reinforcement Learning for Runway Configuration Management: A Literature Review with a Chicago O'Hare (ORD) Case Study

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Abstract

Runway configuration management (RCM) is a safety-critical, high-frequency decision problem at complex multi-runway airports. Focusing on Chicago O'Hare (ORD) as a representative case, this review organizes recent RCM decision-support literature into three families: supervised imitation baselines, conservative offline model-free reinforcement learning (e.g., CQL-based runway configuration assistance), and forecast-/uncertainty-augmented offline RL that incorporates aviation Weather predictions. I synthesize recurring evaluation lenses from prior studies, focusing on decision fidelity to historical operations, feasibility and safety screening, and proxies for operational utility. I also discuss challenges relevant to ORD, including action abstraction under procedural constraints, switching costs, limited coverage of rare regimes, non-stationarity, and forecast error. I conclude with a deployment-oriented agenda for reliable human-in-the-loop RCM decision support.

1 Introduction

RCM mainly focuses on selecting which runway set and direction of operation to use (and when to switch) directly impacts airport capacity, surface congestion, and airborne efficiency, while remaining constrained by safety and procedure. Classic runway-management studies emphasize that the chosen configuration has “significant implications” for capacity and for managing both surface and airborne traffic, and that configuration selection is driven by a mix of constraints (e.g., wind limits) and discretionary considerations (e.g., demand, surface congestion, user preferences), often with limited automation support and short reliable forecasting horizons [Lohr, 2011].

These challenges are amplified at large multi-runway airports with mixed runway geometry and tightly coupled arrival/departure streams. ORD provides a concrete example: it has eight runways, with primary east–West parallels and additional crosswind-oriented runways, leading to non-trivial configuration choices under changing Weather and demand [Chicago Department of Aviation, 2025]. Operationally, ORD alternates between east flow and West flow depending on prevailing winds and Weather. In east flow, departures generally go east and arrivals approach from the West, while the opposite holds in West flow [Chicago Department of Aviation, 2025]. Infrastructure changes further shape feasible and efficient configurations: for instance, Federal Aviation Administration (FAA) capacity documentation notes that new parallel runways such as 10C/28C (commissioned 2013) and 10R/28L (commissioned 2015) are used primarily for arrivals, with 10R/28L playing an especially important role in east-flow operations [Federal Aviation Administration, 2018]. The combination of mixed runway layout, Weather-driven flow reversals, and demand peaks makes RCM a canonical sequential decision-making problem.

From a reinforcement learning (RL) perspective, RCM naturally fits a Markov Decision Process: the “state” aggregates operational and environmental factors (traffic demand, wind and meteorological conditions).

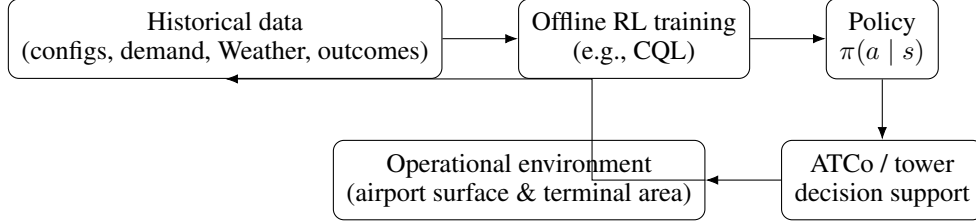


Figure 1: A conceptual view of runway configuration management as an offline RL problem: policies are learned from historical operations data and deployed as decision support for controllers.

logical conditions, time-of-day, surface queues), the “action” selects a runway configuration, and the “reward/utility” encodes performance metrics such as throughput, taxi-out time, and go-arounds [Memarzadeh and Kalyanam, 2025]. However, direct online RL exploration is generally infeasible in safety-critical air traffic operations. This motivates offline RL, where a policy is learned solely from historical data without interacting with the real system. Prior work argues that supervised learning approaches can predict historical controller choices but do not directly optimize an operational utility, and the i.i.d. assumption is violated because each decision influences future states [Memarzadeh et al., 2023a]. Meanwhile, naive off-policy RL can fail under *distributional shift*—when the learned policy differs from the (unknown) behavior policy that generated the dataset—leading to over-optimistic value estimates [Memarzadeh and Kalyanam, 2025, Kumar et al., 2020].

Recent aviation-focused studies therefore adopt conservative offline RL to improve robustness. A representative line of work develops a Runway Configuration Assistance (RCA) tool using Conservative Q-Learning (CQL), leveraging historical data to learn a policy while explicitly regularizing Q-values to remain conservative on out-of-distribution actions [Memarzadeh and Kalyanam, 2025, Kumar et al., 2020]. Survey evidence also highlights RCM as a key air traffic management application area where offline, model-free RL has been deployed to support controller decision-making [Razzaghi et al., 2024]. These developments suggest an emerging research agenda: (i) modeling RCM with operationally meaningful states/actions/rewards, (ii) selecting offline RL algorithms that mitigate distribution shift, and (iii) evaluating policies under safety, interpretability, and real-world deployment constraints.

This literature review focuses on RL methods for runway configuration decisions, with emphasis on offline RL for complex, safety-critical airports (e.g., ORD-like mixed runway geometry and flow reversals). I synthesize how researchers formulate the MDP, construct rewards aligned with operational objectives, address distribution shift and conservatism in offline learning, and evaluate policies against baselines such as controller-preference rules and supervised imitation. The remainder of the paper is organized as follows: Section 2 formalizes RCM as an MDP and summarizes commonly used data sources and metrics; Section 3 reviews supervised and imitation baselines and their limitations; Section 4 surveys offline RL approaches (with a focus on conservative methods such as CQL) and discusses evaluation protocols; Section 5 outlines open challenges including safety assurance, interpretability, and transfer across airports and procedures.

2 Background: RCM at ORD

ORD is a paradigmatic example of complex runway configuration management due to its large runway set, mixed geometry, and frequent flow reversals. ORD currently operates eight runways: 4L/22R, 4R/22L, 9L/27R, 9C/27C, 9R/27L, 10L/28R, 10C/28C, and 10R/28L [Chicago Department of Aviation, 2025]. The primary runway system is oriented east–west (the 09/27 and 10/28 parallels), complemented by crosswind runways (04/22), enabling multiple feasible combinations of arrival and departure runway use [Chicago Department of Aviation, 2025]. This runway richness increases the number of valid “Runway Configuration Plans” (RCPs) and creates strong coupling between arrival and departure streams, which must be coordinated at the runway/procedure level [Lohr, 2011].

East-West flow and weather-driven operations. Runway usage at ORD is primarily dictated by prevailing winds and associated weather conditions. FlyChicago describes two dominant modes: in east flow, aircraft generally depart the primary runways to the east and arrivals approach from the

Table 1: Illustrative ORD runway configuration examples in Visual conditions (subset) from the FAA capacity profile.

Flow & Scenario	Arrival runways	Departure runways	Throughput (ops/hr)
East (current)	09L, 10C, 10R	09R, 10L	214–223
East (Quad Arrivals)	09C, 09L, 10C, 10R	09R, 10L	292 (model est.)
West (current)	27L, 27R, 28C	22L, 28R	214–224
West (Quad Arrivals)	27C, 27R, 28C, 28L	27L, 28R	292 (model est.)

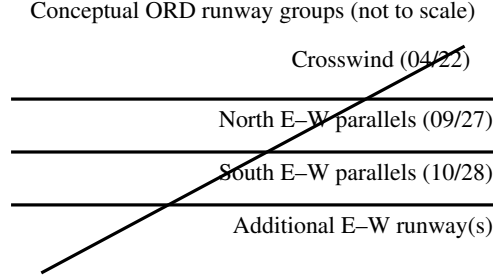


Figure 2: ORD features multiple parallel east–west runways plus crosswind runways, enabling many runway configuration plans. Flow direction (east vs. west) and Weather regimes strongly influence which configuration is feasible and efficient.

west; in west flow, departures generally go to the west and arrivals come from the east [Chicago Department of Aviation, 2025]. Flow changes can persist for extended periods but may switch when winds and weather evolve, or when ground conditions such as runway closures/maintenance require operational adjustments [Chicago Department of Aviation, 2025].

The FAA capacity profile formalizes the operational context by stratifying conditions into Visual, Marginal, and Instrument regimes based on ceiling and visibility thresholds (e.g., Visual: at least 5500 ft ceiling and 7 miles visibility; Instrument: below 1000 ft ceiling or 3 miles visibility) [Federal Aviation Administration, 2018]. These regimes matter because feasible arrival/departure rates and preferred runway sets depend on wake separation rules, approach procedures, and runway interaction constraints under degraded visibility [Federal Aviation Administration, 2018].

Representative configurations and capacity trade-offs. For high-demand periods, the FAA profile reports representative arrival/departure runway sets and associated hourly throughput. Table 1 summarizes a subset of commonly referenced Visual-condition configurations in both east and west flow. A key operational insight is that increasing arrival capacity (e.g., “Quad Arrivals”) may reduce departure capacity or impose additional separation constraints, illustrating the multi-objective nature of configuration choice [Federal Aviation Administration, 2018]. The profile also notes operational nuances such as “departure fanning” using Runway 22L in some cases, and constraints when certain arrival runways are active, which directly affect what configurations are practical at a given time [Federal Aviation Administration, 2018].

Finally, it is worth emphasizing that RCM is not only about choosing a configuration, but also about timing the switch. Configuration changes can be disruptive and require coordination, and in practice operators must balance anticipated near-term traffic and weather evolution against the transition cost and risk of switching too early or too late [Lohr, 2011, Federal Aviation Administration, 2018].

3 Problem Formulation

I model runway configuration management as a discrete-time sequential decision problem, aligning with prior offline RL formulations for runway configuration assistance [Memarzadeh and Kalyanam, 2025]. Decisions are made at regular epochs indexed by $t \in \{0, 1, 2, \dots\}$; following Memarzadeh and Kalyanam [2025], a natural granularity is $\Delta t = 15$ minutes, though other operationally meaningful intervals (e.g., 30 minutes) are possible.

MDP definition. Let the problem be an MDP with tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, U)$. At time t , the decision maker (agent) observes a state $s_t \in \mathcal{S}$ and selects an action $a_t \in \mathcal{A}$, where a_t corresponds to a runway configuration plan (arrival/departure runway set and direction of operation). The system transitions to $s_{t+1} \sim \mathcal{P}(\cdot \mid s_t, a_t)$ and yields a scalar utility (reward) $u_t = U(s_t, a_t, s_{t+1})$.

State space. The state aggregates variables that influence configuration choice, including: (i) Traffic demand and imbalance (e.g., arrival/departure demand rates, backlog proxies), (ii) Meteorology (e.g., surface wind speed/direction, ceiling, visibility; optionally forecasts), (iii) Time context (time-of-day, day-of-week/seasonality), and (iv) Operational context, such as the currently active configuration and a measure of “time since last switch” (to capture transition inertia) [Memarzadeh and Kalyanam, 2025, Lohr, 2011]. FAA capacity profiling also indicates that configuration usage depends on Weather regimes and other operational factors, and that configuration data can be obtained from FAA ASPM and related sources [Federal Aviation Administration, 2018].

Action space. I assume a finite, discrete action set $\mathcal{A}$ consisting of valid runway configurations observed historically or enumerated from operationally plausible plans. For ORD, an action can be represented as a labeled plan, e.g., East-current, East-Quad-Arrivals, West-current, West-Quad-Arrivals, each mapping to an arrival/departure runway set (Table 1) [Federal Aviation Administration, 2018]. To ensure operational realism, actions can be masked when they violate basic constraints (e.g., tailwind/crosswind risk) or when runways are unavailable due to closures/maintenance [Memarzadeh and Kalyanam, 2025, Chicago Department of Aviation, 2025].

Utility function (reward). Following the RCA line of work, I use a weighted multi-objective utility that captures operational goals such as maximizing throughput, minimizing surface transit times (taxi-out/taxi-in), and mitigating safety-related events such as go-arounds [Memarzadeh and Kalyanam, 2025]. To capture the practical cost of changing configurations, I incorporate a switching penalty. A simple instantiation is:

$$u_t = w_{thr} \cdot Throughput(t) - w_{taxi} \cdot TaxiTime(t) - w_{ga} \cdot GoArouns(t) - w_{sw} \cdot \mathbb{I}[a_t \neq a_{t-1}], \quad (1)$$

where $w_{\cdot} \geq 0$ are tunable weights (often elicited with subject matter experts) and $\mathbb{I}[\cdot]$ is an indicator. More refined utilities may include explicit arrival/departure fairness terms, queue stability, or constraint-violation penalties [Memarzadeh and Kalyanam, 2025, Lohr, 2011].

Offline RL objective and data. Because online exploration is not acceptable in safety-critical operations, I assume access to an offline dataset $\mathcal{D} = \{(s_t, a_t, u_t, s_{t+1})\}$ generated by historical controller decisions (a behavior policy) [Memarzadeh and Kalyanam, 2025, Memarzadeh et al., 2023a]. The learning objective is to find a policy $\pi(a \mid s)$ maximizing the expected discounted return:

$$J(\pi) = \mathbb{E}_{\pi, \mathcal{P}} \left[\sum_{t=0}^{\infty} \gamma^t u_t \right], \quad 0 < \gamma < 1, \quad (2)$$

while mitigating the fundamental challenge of offline RL: distributional shift between the learned policy and the data-collection

4 Method 1: Supervised Prediction Baselines

A natural first step for RCM is to treat the task as a supervised prediction problem: given information available at time t (traffic, Weather, operational context), predict the runway configuration used at the next decision epoch $t+1$. This approach is often framed as imitation learning / behavioral cloning of historical controller decisions, and has shown strong predictive accuracy when trained on large-scale historical data. [Puranik et al., 2023]

4.1 Problem setup as next-step classification

Let x_t denote the feature vector available at time t , including (but not limited to) surface wind speed/direction, visibility/ceiling, time-of-day, demand proxies (e.g., scheduled arrivals/departures), and the incumbent configuration. Prior work shows these factors are central drivers of configuration

Algorithm 1 Supervised next-step RCM prediction (behavioral cloning)

Require: $\mathcal{D} = \{(x_t, y_{t+1})\}$, model $p_\phi(a \mid x)$, lr η , epochs E

Ensure: ϕ^*

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1:  $\phi \leftarrow \text{Init}()$ 
2: for  $e = 1, \dots, E$  do
3:    $\mathcal{B} \sim \text{MiniBatch}(\mathcal{D})$ 
4:    $\ell(\phi; \mathcal{B}) \leftarrow - \sum_{(x,y) \in \mathcal{B}} \log p_\phi(y \mid x)$ 
5:    $\phi \leftarrow \phi - \eta \nabla_\phi \ell(\phi; \mathcal{B})$ 
6: end for
7:  $\phi^* \leftarrow \text{SelectByCV}(\phi)$  (metric : wF1)
8: Report Acc, wF1, ConfMat, OV on test
9: return  $\phi^*$ 
```

▷ cross-entropy

choice, and explicitly models them at 15-minute resolution for multi-hour horizons. [Avery and Balakrishnan, 2016, Puranik et al., 2023]

I define the label $y_{t+1} \in \mathcal{A}$ as the runway configuration active in the next interval (e.g., next 15 minutes). The supervised baseline learns a parametric classifier f_ϕ producing $p_\phi(a \mid x_t)$ over $a \in \mathcal{A}$. Training minimizes a standard cross-entropy objective:

$$\mathcal{L}(\phi) = -\frac{1}{|\mathcal{D}|} \sum_{(x_t, y_{t+1}) \in \mathcal{D}} \log p_\phi(y_{t+1} \mid x_t). \quad (3)$$

Because runway configurations are typically *imbalanced* (some configurations are used much more frequently), model selection can benefit from imbalance-aware metrics (e.g., weighted F1) rather than plain accuracy. [Puranik et al., 2023]

4.2 Baseline models

I introduce several supervised baselines with increasing modeling capacity:

B0: Persistence / inertia baseline. Predict that the next configuration remains unchanged: $\hat{y}_{t+1} = y_t$. Empirically, “operational inertia” is strong because switching requires coordination and can reduce throughput, so this baseline is often surprisingly competitive. [Avery and Balakrishnan, 2016]

B1: Wind-only rule baseline. Use a coarse rule based on wind direction (dominant driver in many airports) to choose the feasible flow and map to a preferred configuration for that flow. This mirrors the fact that wind is often a top explanatory factor in data-driven models. [Puranik et al., 2023]

B2: Multinomial logistic regression. A probabilistic model that predicts a distribution over runway combinations using a compact meteorological feature set (wind, visibility, daylight, number of runways in use), often used for runway-use forecasting with uncertainty quantification. [de Leege et al., 2016]

B3: Tree-based classifiers (Random Forest / Gradient Boosting). More expressive non-linear models that capture feature interactions. In a large-scale supervised RCM study that fused multiple data sources at 15-minute granularity, Random Forest outperformed other considered classifiers (e.g., SVM, Gradient Boosting) and achieved test accuracies around 93% (CLT) and 73% (DEN), highlighting both the promise and the airport-complexity dependence. [Puranik et al., 2023]

4.3 Training pipeline and evaluation

Algorithm 1 summarizes the end-to-end supervised pipeline used for our baselines. A practical note is that constructing reliable labels (i.e., extracting the actual runway configuration in use) can be a non-trivial and time-consuming step, and may require airport-specific offline work. [Puranik et al., 2023]

Metrics. Beyond top-1 accuracy and weighted F1, I report (i) confusion matrices to diagnose which configurations are systematically confused (often among very similar configurations), and (ii) a safety-relevance “obvious violation” metric that counts how often the model recommends a clearly unreasonable configuration under strong winds (as defined with SME rules). [Puranik et al., 2023]

4.4 Why these are only baselines

While supervised prediction can be highly accurate at imitating historical choices, it does *not* optimize an explicit operational utility (e.g., taxi time, throughput, go-arounds) and therefore cannot directly correct inefficiencies in historical decision-making. This limitation is a key motivation for offline RL methods in later sections. [Memarzadeh et al., 2023b]

5 Method 2: Offline Model-Free RL for RCM

Supervised prediction baselines imitate historical decisions but do not directly optimize operational outcomes. In contrast, the Runway Configuration Assistance (RCA) line of work formulates runway configuration management (RCM) as an *offline* sequential decision problem, where a policy is learned solely from historical data to optimize a weighted multi-objective utility (e.g., throughput, surface transit times, and go-arounds). [Memarzadeh and Kalyanam, 2025] This is essential in aviation settings where online exploration is not acceptable and accurate simulators may be costly or unavailable. [Memarzadeh et al., 2023a]

5.1 Offline value-based RL for discrete runway configurations

Let $\mathcal{D} = \{(s_t, a_t, u_t, s_{t+1})\}$ be the logged dataset collected under an unknown behavior policy π_β . For discrete action sets (finite runway configuration plans), a common model-free approach is Q-learning with a function approximator $Q_\theta(s, a)$. A brute-force offline adaptation of Deep Q-Network (DQN) minimizes the Bellman error: $y_t = u_t + \gamma \max_{a' \in \mathcal{A}(s_{t+1})} Q_{\bar{\theta}}(s_{t+1}, a')$,

$\mathcal{L}_{DQN}(\theta) = \mathbb{E}_{(s,a,u,s') \sim \mathcal{D}} \left[(Q_\theta(s, a) - y)^2 \right]$, where $\bar{\theta}$ denotes a target network and $\mathcal{A}(s)$ is an (optional) feasibility-masked action set (e.g., remove clearly infeasible configurations under severe winds/closures). However, in purely offline settings, standard off-policy Q-learning can overestimate values due to distributional shift and out-of-distribution (OOD) actions. [Kumar et al., 2020, Memarzadeh et al., 2023a]

5.2 Conservative Q-Learning (CQL) for mitigating distributional shift

RCA adopts Conservative Q-Learning (CQL), which augments Bellman error minimization with a conservative regularizer that penalizes high Q-values on actions not supported by the dataset. [Memarzadeh and Kalyanam, 2025, Kumar et al., 2020] For discrete actions, a widely used instantiation is:

$$\mathcal{L}_{CQL}(\theta) = \mathcal{L}_{DQN}(\theta) + \alpha \mathbb{E}_{s \sim \mathcal{D}} \left[\log \sum_{a \in \mathcal{A}(s)} \exp(Q_\theta(s, a)) - \mathbb{E}_{a \sim \mathcal{D}(\cdot|s)} [Q_\theta(s, a)] \right], \quad (4)$$

where $\alpha \geq 0$ controls conservatism. Intuitively, the $\log \sum \exp$ term pushes down the values of actions broadly (including OOD actions), while the data-action expectation counteracts excessive pessimism on in-distribution actions. [Kumar et al., 2020] In RCA, CQL is used to learn a policy that encourages good historical decisions and avoids poor ones, guided by a weighted multi-objective utility, and compared against baselines such as “most frequent configuration”, supervised learning, and offline DQN. [Memarzadeh and Kalyanam, 2025]

5.3 RCA training and decision-support pipeline

Algorithm 2 summarizes the offline CQL training loop for RCM with discrete configuration actions. At inference time, the tool outputs a ranked list of recommended configurations (or a top-1 recommendation) to support controllers, rather than autonomously executing changes. [Memarzadeh and Kalyanam, 2025]

Algorithm 2 Offline CQL for runway configuration management (discrete actions)

Require: Dataset $\mathcal{D} = \{(s, a, u, s')\}$, action set $\mathcal{A}$, discount γ , lr η , CQL light α , steps K

Ensure: Q-network parameters θ

```
1: Initialize  $Q_\theta$ , target network  $Q_{\bar{\theta}} \leftarrow Q_\theta$ 
2: for  $k = 1, \dots, K$  do
3:   Sample minibatch  $\mathcal{B} \subset \mathcal{D}$ 
4:   for all  $(s, a, u, s') \in \mathcal{B}$  do
5:      $\mathcal{A}(s') \leftarrow \text{Mask}(\mathcal{A}, s')$  ▷ optional feasibility mask
6:      $y \leftarrow u + \gamma \max_{a' \in \mathcal{A}(s')} Q_{\bar{\theta}}(s', a')$ 
7:   end for
8:    $\mathcal{L}_{DQN} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{(s, a, \cdot, \cdot) \in \mathcal{B}} (Q_\theta(s, a) - y)^2$ 
9:    $\mathcal{R}_{CQL} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{s \in \mathcal{B}} \left( \log \sum_{a \in \mathcal{A}(s)} e^{Q_\theta(s, a)} - Q_\theta(s, a_{\mathcal{D}}) \right)$  ▷  $a_{\mathcal{D}}$  is the dataset action paired with  $s$  in  $\mathcal{B}$ 
10:   $\theta \leftarrow \theta - \eta \nabla_\theta (\mathcal{L}_{DQN} + \alpha \mathcal{R}_{CQL})$ 
11:  Periodically update target:  $\bar{\theta} \leftarrow \theta$ 
12: end for
13: return  $\theta$ 
```

Strengths. Compared with imitation-style supervised learning, offline RL optimizes an explicit utility aligned with operational goals, and can, in principle, recommend actions that are *better than average historical behavior* under similar conditions. RCA further improves robustness by using CQL to reduce overestimation under distributional shift. [Memarzadeh and Kalyanam, 2025, Kumar et al., 2020]

Limitations and open issues. Despite conservatism, offline RL is fundamentally limited by dataset coverage: rare Weather regimes, unusual traffic patterns, or configurations seldom used historically can remain hard to evaluate, and model selection/offline policy evaluation (OPE) remains non-trivial. RCA therefore emphasizes evaluation via operational metrics and comparisons against multiple baselines, but broader deployment requires stronger assurance (e.g., constraint checking, verification, and human-in-the-loop protocols). [Memarzadeh and Kalyanam, 2025, Memarzadeh et al., 2023a]

6 Method 3: Forecast- and Uncertainty-Augmented Offline RL

A key limitation of purely reactive RCM policies is that configuration switches are often triggered by (anticipated) near-term changes in wind and ceiling/visibility, not only by the current snapshot. Motivated by this, recent work augments offline RL for RCM with *explicit aviation Weather forecasts* and studies how forecast information improves both recommendation accuracy and adaptability to fast wind shifts. [Nethi et al., 2024]

6.1 Forecast augmentation: expanding the state with short-horizon predictions

Let s_t denote the “current-condition” state as in Section 3 (demand, current METAR-like Weather, time context, incumbent configuration, etc.). Forecast augmentation appends a forecast vector f_t that summarizes predicted future meteorology over a short horizon:

$$\tilde{s}_t = [s_t; f_t], \quad f_t = \{\hat{w}_{t+h}, \hat{v}_{t+h}, \hat{c}_{t+h}\}_{h \in \mathcal{H}}, \quad (5)$$

where $\hat{w}_{t+h}$ may represent forecast wind direction/speed, $\hat{v}_{t+h}$ visibility, and $\hat{c}_{t+h}$ ceiling (or categorical VFR/IFR indicators) at lead times $h \in \mathcal{H}$ (e.g., $h \in \{1, 2, 3, 4\}$ steps of 15 minutes).

In the forecast-augmented RCA framework, the forecast sources include LAMP and TAF products, and adding these forecast variables improved the decision-support tool’s prediction accuracy and responsiveness to quick wind changes. To address scalability issues from adding discrete forecast variables, the approach transitions to a continuous state-space representation while keeping runway configurations as discrete actions. [Nethi et al., 2024]

Algorithm 3 Forecast- and uncertainty-augmented offline RL for RCM (discrete actions)

Require: Logged data $\mathcal{D} = \{(s, a, u, s')\}$, forecast provider $\text{FORECAST}(\cdot)$, horizons $\mathcal{H}$, discount γ , lr η , CQL weight α , uncertainty weight β

Ensure: Q-network $Q_\theta(\tilde{s}, a)$ and decision-support policy π

Offline training

- 1: Build augmented tuples $\tilde{\mathcal{D}} = \{(\tilde{s}, a, u, \tilde{s}')\}$ using Eq. eq:augstate
- 2: $k = 1, \dots, K$
- 3: Sample minibatch $\mathcal{B} \subset \tilde{\mathcal{D}}$
- 4: **for all** $(\tilde{s}, a, u, \tilde{s}') \in \mathcal{B}$ **do**
- 5: $y \leftarrow u + \gamma \max_{a' \in \mathcal{A}(\tilde{s}')} Q_{\tilde{\theta}}(\tilde{s}', a')$
- 6: **end for**
- 7: $\mathcal{L}_{Bell} \leftarrow \frac{1}{|\mathcal{B}|} \sum (Q_\theta(\tilde{s}, a) - y)^2$
- 8: $\mathcal{R}_{CQL} \leftarrow \frac{1}{|\mathcal{B}|} \sum \left(\log \sum_{a' \in \mathcal{A}(\tilde{s})} e^{Q_\theta(\tilde{s}, a')} - Q_\theta(\tilde{s}, a) \right)$
- 9: $\theta \leftarrow \theta - \eta \nabla_\theta (\mathcal{L}_{Bell} + \alpha \mathcal{R}_{CQL})$
- 10: Periodically update target: $\tilde{\theta} \leftarrow \theta$
- 11:

Online recommendation (decision support)

- 12: Observe current s_t and obtain forecast scenarios $\{f_t^{(m)}\}_{m=1}^M \leftarrow \text{FORECAST}(t, \mathcal{H})$
 - 13: **for all** $a \in \mathcal{A}(\tilde{s}_t)$ **do**
 - 14: $\hat{Q}(a) \leftarrow \min_{m \in [M]} Q_\theta([s_t; f_t^{(m)}], a)$ ▷ robust to forecast error
 - 15: $\sigma(a) \leftarrow \text{UNC}([s_t; f_t^{(1:M)}], a)$ ▷ e.g., ensemble / Q-distribution
 - 16: $\text{SCORE}(a) \leftarrow \hat{Q}(a) - \beta \sigma(a)$ ▷ pessimistic action scoring
 - 17: **end for**
 - 18: $\pi(\tilde{s}_t) \leftarrow \arg \max_a \text{SCORE}(a)$
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6.2 Uncertainty: forecasts are imperfect and offline RL is brittle

Two uncertainties matter in practice:

(i) Forecast uncertainty. Aviation forecast products provide frequent updates and are widely used for situational awareness, but they are not perfect. For example, FAA guidance describes LAMP as an automated product updated hourly with forecasts reaching to the next day, and recommends using it in combination with TAF and other tools rather than treating it as a single source of truth. [Federal Aviation Administration, 2017] This implies that RCM policies should be robust to forecast errors and regime shifts.

(ii) Offline-RL uncertainty / OOD actions. Offline RL can overestimate values for actions not in the supported by the logged dataset (distribution shift), which motivates both conservative objectives (e.g., CQL) and *uncertainty-aware* pessimism that penalizes risky actions when uncertainty is high. For example, QDQ proposes penalizing Q-values in OOD regions using uncertainty estimated from a learned Q-distribution, and RORL targets robustness under observation deviations. [Zhang et al., 2024, Yang et al., 2022]

6.3 A practical recipe: forecast-augmented CQL with robust / uncertainty-aware action selection

Algorithm 3 summarizes a pragmatic method that (a) trains a conservative offline Q-function on the augmented state $\tilde{s}_t$ and (b) selects actions using a lower-confidence-style score. Forecast uncertainty is handled by sampling multiple forecast scenarios (e.g., from probabilistic forecasts, ensembles, or perturbations), then using a robust aggregation (worst-case or average). Offline-RL uncertainty is handled by penalizing actions with high estimated uncertainty.

Evaluation notes. For aviation deployment, evaluation typically includes both (i) agreement with historical controller selections and (ii) “obvious violation” checks (e.g., recommendations that violate clear wind constraints), alongside operational metrics derived from logs. These metrics have been

Table 2: High-level comparison of RCM method families discussed in Sec. 4–6 (literature-reported evaluation lenses).

Family	What it learns	Typical strengths	Typical ORD-relevant risks
Supervised imitation [Puranik et al., 2023, Avery and Balakrishnan, 2016, de Leege et al., 2016]	$p(a_{t+1} \mid s_t)$ (behavioral cloning)	Strong decision fidelity; simple training/evaluation; interpretable feature effects	Does not optimize utility; can reproduce suboptimal inertia or rare-regime errors; sensitive to class imbalance
Offline model-free RL (CQL/RCA) [Memarzadeh and Kalyanam, 2025]	$Q(s, a)$ / policy maximizing multi-objective utility	Explicitly optimizes utility; can encode switching costs; conservative variants reduce risky extrapolation	Limited by dataset coverage (rare Weather/IRROPS); requires strong feasibility screening and careful selection
Forecast-/uncertainty-augmented offline RL [Nethi et al., 2024, Federal Aviation Administration, 2017]	$Q([s; f], a)$ with short-horizon forecasts	Anticipatory decisions for rapid wind shifts; reduces myopic switching	Forecast error introduces another failure mode; robustness requires conservative/robust decision rules

used to validate RCA-style tools across multiple airports with differing complexity. [Kalyanam and Tejasen, 2024]

7 Discussion: Toward Reliable RCM for Complex Multi-Runway Airports

Following the example literature review structure, I consolidate “evaluation” into the discussion: I first compare the reviewed approaches along a small set of evaluation lenses reported in the literature, and then summarize open challenges that remain salient for complex multi-runway airports such as ORD.

7.1 Comparison and evaluation lenses in the RCM literature

Across supervised RCM prediction and offline-RL-based RCA-style decision support, prior work typically assesses methods along three recurring lenses: (i) *decision fidelity* (agreement with historical configurations), (ii) *safety/feasibility sanity checks* (e.g., rule-based screening under strong winds), and (iii) *operational utility proxies* aligned with the reward design (throughput, surface time, or other outcome KPIs derived from logs). These lenses appear naturally because RCM is human-in-the-loop and safety critical, while the logged data distribution is highly imbalanced across configurations. [Puranik et al., 2023, Memarzadeh and Kalyanam, 2025]

Table 2 summarizes how the three method families (Sec. 4–6) position themselves relative to these lenses for an ORD-like setting.

Overall, the literature suggests that supervised models are strong baselines for *fidelity*, while offline RL is motivated when one wants to optimize a defined operational utility (including explicit switch penalties). Forecast augmentation is promising for fast-changing conditions, but reliability hinges on how forecast uncertainty is handled rather than on adding forecasts alone.

7.2 Challenges and limitations for reliable ORD-scale RCM

Despite encouraging results reported in prior work, several challenges remain particularly important for complex multi-runway airports:

Action abstraction and feasibility constraints. For ORD-like airports, the configuration space is large but structured. A practical action library must reflect procedure feasibility (closures, wind components, arrival/departure coupling), not merely “seen historically”, otherwise both imitation and offline RL can generate infeasible recommendations.

Switching as a first-class objective. RCM is not only about picking a high-capacity configuration; it is about *when to switch*. The literature implicitly addresses this via inertia in data (supervised) or via switch penalties (offline RL), but ORD-scale reliability likely benefits from explicit temporal abstractions (e.g., flow-level decisions vs within-flow choices).

Rare regimes and non-stationarity. Offline RL is fundamentally constrained by what the dataset covers. Rare low-visibility regimes or atypical crosswinds may be underrepresented, yet are exactly when decision support is most needed. In addition, procedure updates and infrastructure changes can shift the state–outcome relationship over time, motivating time-split validation and periodic re-training rather than a one-time model build.

Forecast uncertainty and trust calibration. Forecast augmentation can help anticipatory decisions, but aviation forecast products are imperfect and updated frequently. Decision support should therefore communicate uncertainty (or provide top- k ranked options) to reduce automation bias, especially under rapidly changing wind conditions. [Federal Aviation Administration, 2017, Nethi et al., 2024]

These challenges suggest that “reliable RCM” at complex airports is less about a single algorithmic win, and more about combining conservative learning, airport-specific feasibility logic, and human-centered decision support.

8 Conclusion

This review surveyed runway configuration management (RCM) through the lens of decision support for complex multi-runway airports (e.g., ORD), and organized the literature into three method families: supervised imitation baselines, offline model-free RL with conservative learning (e.g., CQL-based RCA), and forecast-/uncertainty-augmented offline RL. Across these works, a consistent lesson is that high predictive agreement alone is insufficient: reliable RCM requires explicit utility optimization, conservatism against distribution shift, and deployment-grade guardrails (constraint screening, conservative OPE, and human-in-the-loop operation). Looking forward, progress toward robust ORD-scale deployment will likely depend on:

- **Constraint-aware learning:** incorporate hard procedure/wind/availability constraints beyond post-hoc masking;
- **Uncertainty-aware selection:** improve confidence-bound OPE and robust decision rules under rare regimes;
- **Sustained operations:** establish monitoring, drift detection, and periodic re-training/re-certification as airport infrastructure and procedures evolve.

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